

Worksheet 3
MTH 203 (Multivariate Calculus)
Date: 01-09-2026

Q1. Suppose $f(x, y) = 5x^2 + y^2$. Using the definition of partial derivatives, compute f_x and f_y . Also, evaluate $f_x(2, 3)$ and $f_y(2, 3)$.

Q2. Show that $f_x(0, 0) = 0$, but $f_y(0, 0)$ does not exist, where

$$f(x, y) = \begin{cases} (x^2 + y) \sin\left(\frac{1}{x^2 + y^2}\right), & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

Provide a detailed analysis.

Q3. Find $\frac{\partial z}{\partial x}$ if the equation

$$yz - \ln z = x + y$$

defines z as a function of two independent variables x and y and the partial derivative exists.

Q4. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}$ be a given linear transformation. Compute $\frac{\partial T}{\partial x}$, $\frac{\partial T}{\partial y}$ and $\frac{\partial T}{\partial z}$ at the point (x_0, y_0, z_0) .

Q5. Verify that

$$T(x, t) = e^{-t} \cos\left(\frac{x}{c}\right)$$

satisfies the heat equation

$$\frac{\partial T}{\partial t} = c^2 \frac{\partial^2 T}{\partial x^2}.$$

Q6. By direct calculation, show that

$$f_{xyz} = f_{yzx} = f_{zyx}$$

for the function

$$f(x, y, z) = xyz + x^2 y^3 z^4.$$

Q7. Define differentiability of a function $f : D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$ in a rectangular domain D . Further, show that if

$$f(x, y)$$

is differentiable at (x_0, y_0) , then it is also continuous at (x_0, y_0) .

Q8. Given

$$z = u(x, y)e^{ax+by}$$

and

$$\frac{\partial^2 u}{\partial x \partial y} = 0,$$

find the values of the constants a and b such that

$$\frac{\partial^2 z}{\partial x \partial y} - \frac{\partial z}{\partial x} - \frac{\partial z}{\partial y} + z = 0.$$